

SANJAVAN GHODASARA

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EDUCATION

Stevens Institute of Technology

Master of Science in Machine Learning

Coursework: Deep Learning (A), Machine Learning, Artificial Intelligence, Natural Language Processing, Pattern Recognition

Silver Oak College of Engineering & Technology, Silver Oak University

Bachelor of Technology in Information Technology

Hoboken, NJ

Expected May 2027

Ahmedabad, India

May 2025

RESEARCH EXPERIENCE

Graduate Research Assistant – School of Computing

Oct 2025 – Present

Stevens Institute of Technology · Advisor: Prof. Tiffany Wenting Li

Hoboken, NJ

- Studying how students learn with an AI chatbot, reading and web search (919 coded messages, 344 sessions); CHI 2027 poster and L@S 2027 paper in preparation
- Wrote a multivariate Poisson mixture model from scratch (BIC/ICL) and Jensen–Shannon k-medoids clustering to profile learner behavior
- Built the analysis pipeline for all 4 research questions, incl. a custom FISTA elastic-net solver; co-coded the corpus with my advisor to 100% agreement

PROFESSIONAL EXPERIENCE

AI/ML Engg. – Network Automation

Jan 2025 – May 2025

Reliance Jio

Navi Mumbai, India

- Fine-tuned VGG19 (TensorFlow) on EuroSAT to classify satellite tiles for 5G/6G site planning at 93% accuracy, with human-in-the-loop QGIS review
- Built geospatial ETL for 10 cities in PostGIS and MongoDB, validated predictions against OpenStreetMap, and fed clutter layers into Atoll RF simulation
- Deployed batch inference on GCP (Docker, GKE): 50K tiles in 22 hours (≈ 1.6 s per tile), cutting manual GIS effort by an estimated 70%

AI/ML Team Lead – IBM India Software Lab (WatsonX)

May 2024 – Sep 2024

IBM India · also Data Analyst, Government Infrastructure Analytics (Jul – Sep 2024)

Ahmedabad, India

- Led a 6-person student team building a BERT/GPT healthcare chatbot on IBM Watson; built NHA1 road-safety dashboards (Python, SQL, Tableau)

Junior Software Developer

Aug 2022 – Feb 2023

LTIMindtree

Mumbai, India

- Built backend microservices in FastAPI and cross-platform mobile apps in Flutter

KEY PROJECTS

Synovia – Surgical Outcome Prediction · Best Healthcare Hack, HackPrinceton Fall 2025 · Covered by [McMaster News](#) [Devpost](#) | [GitHub](#)

Predicts neurological deficits before brain surgery by combining finite-element simulation with Gemini multimodal reasoning; led the ML side and built the simulation API in a 3-person team.

Synaptix – AI Drug Repurposing · [MLH] Best Use of Vultr, Hacklytics 2026 (Georgia Tech)

[Devpost](#) | [GitHub](#) | [Live](#)

Ranks FDA-approved drugs for untreated diseases using TransE embeddings on the DRKG knowledge graph (97K entities, 5.8M triples) blended with RDKit molecular similarity, with path-based explanations; built the ML engine and React frontend.

Night Owl – NYC Rat Surveillance · DivHacks 2026 (Columbia) (Pitching to NYC Mayor office)

[Devpost](#) | [GitHub](#)

Built the full modeling stack (H3 cell-by-month panel, LightGBM, a “silence score”, rolling backtest, sensor-site optimizer) to find rat activity that 311 complaints miss; over a 119-month backtest its picks found rat signs at 19.5% of swept lots vs. 15.2% for the 311 baseline.

GeneAI – Pharmacogenomics Decision Tool · Top 7, HackVoyagers pitch, HackIllinois 2026

[Devpost](#) | [GitHub](#)

Predicts adverse drug reactions from a patient’s genes instead of population averages; built the data pipeline (CPIC, PubChem, DrugBank → risk labels), FastAPI service and deployment.

HONORS & LEADERSHIP

- Hackathon prizes:** Best Healthcare Hack, HackPrinceton Fall 2025 | [MLH] Best Use of Vultr, Hacklytics 2026 | Best Use of Flowglad, Columbia DevFest 2026 (ContractPilot) | Top 7 pitch, HackIllinois 2026 | 8+ overall Hackathon wins | 18+ Hackathons & events
- Selected:** Judge, HackHarvard 2026 | Invite-only DevHouse SF residency for student founders (Dec 2025)
- Leadership:** Captain, Gujarat State football team (2021 Cup of India) | Team Leader, TEDx Silver Oak University | Technical Team, IEEE SB

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

ML & AI: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face, LightGBM, KG embeddings, CLIP, FAISS, Gemini API, RAG, LLM evaluation

Statistics: Mixture models (Poisson, GMM/LPA), k-medoids, sequential pattern mining (BIDE), elastic net (custom FISTA), BIC/ICL

Geospatial: PostGIS, GeoPandas, Shapely, Rasterio, QGIS, OpenStreetMap, H3, Atoll

Engineering & Cloud: FastAPI, React, Next.js, Docker, Kubernetes, GCP, Vercel, Supabase, PostgreSQL, MongoDB, GitHub Actions